

LLD Case Study 2: Library Management System

Target Patterns: Observer Pattern, State Pattern, Association vs Aggregation

Key Engineering Focus: Domain modeling, book item state transitions, reservation queues, and fine calculations.

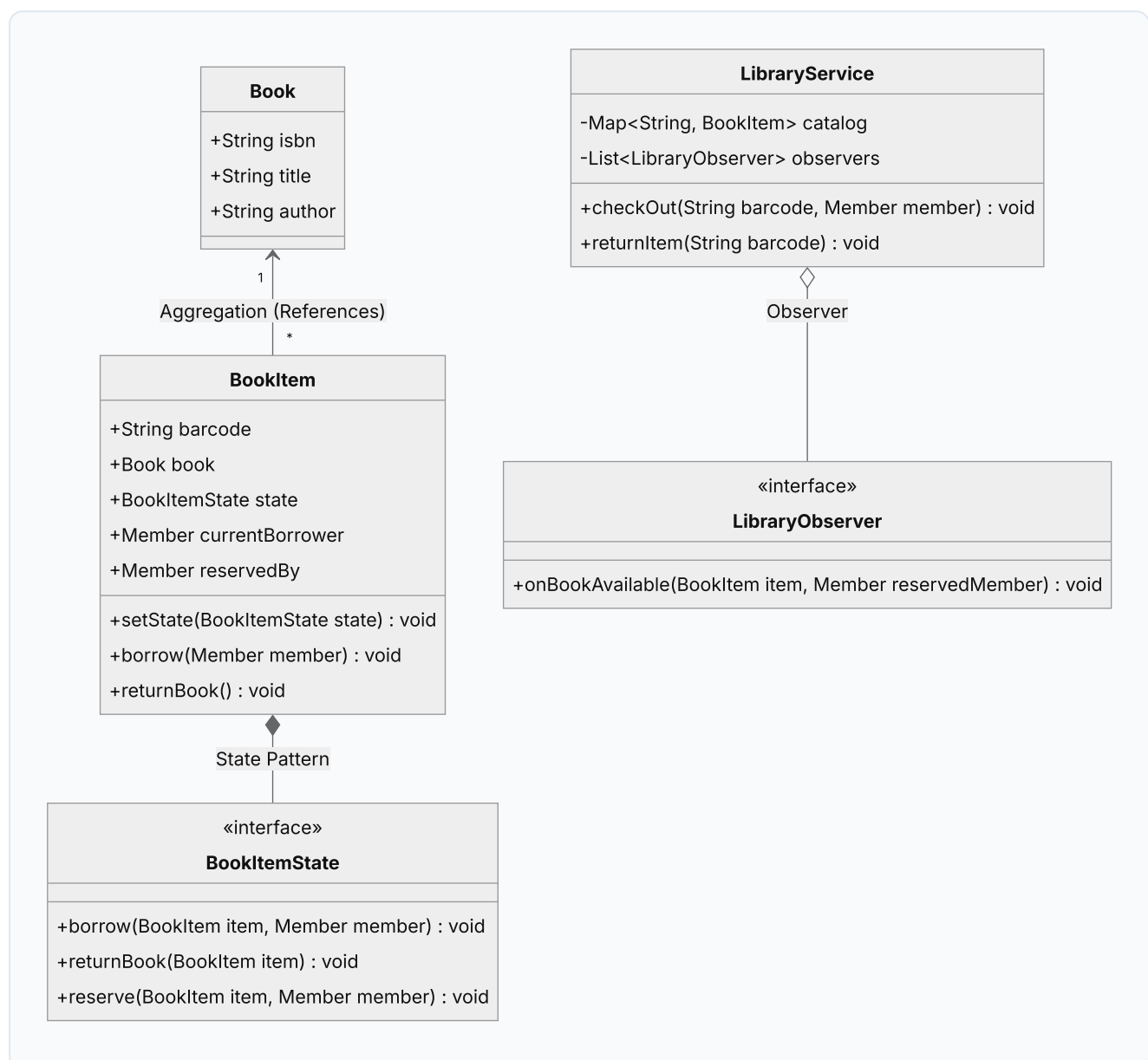
1. Problem Statement & Functional Requirements

Design an enterprise Library Management System capable of cataloging books, managing physical copies (BookItems), issuing loans to members, tracking overdue fines, and managing waitlists for popular books.

Key Requirements:

- 1. Book vs BookItem (Aggregation/Composition):** A `Book` represents metadata (Title, ISBN, Author). A `BookItem` represents a physical barcoded copy on a shelf.
 - 2. State Pattern:** A `BookItem` transitions through states: `Available` → `Issued` → `Reserved` → `Lost`.
 - 3. Observer Pattern:** When an issued book that has an active reservation is returned, notify the reserved member automatically.
 - 4. Fines & Limits:** A member can borrow at most N books. Overdue returns accumulate daily fines.
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2. Architecture & UML Class Diagram



3. Production-Grade Python Implementation